

Whole-Brain Intracellular pH Mapping of Gliomas Using High-Resolution ^{31}P MR Spectroscopic Imaging at 7.0 T

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Malignant tumors commonly exhibit a reversed pH gradient compared with normal tissue, with a more acidic extracellular pH and an alkaline intracellular pH (pH_i). In this prospective study, pH_i values in gliomas were quantified using high-resolution phosphorous ^{31}P spectroscopic MRI at 7.0 T and were used to correlate pH_i alterations with histopathologic findings. A total of 12 participants (mean age, 58 years $\pm$ 18 [SD]; seven male, five female) with histopathologically proven, newly diagnosed glioma were included between September 2018 and November 2019. The ^{31}P spectroscopic MRI scans were acquired using a double-resonant $^{31}\text{P}/^1\text{H}$ phased-array head coil together with a three-dimensional (3D) ^{31}P chemical shift imaging sequence (5.7-mL voxel volume) performed with a 7.0-T whole-body system. The 3D volumetric segmentations were performed for the whole-tumor volumes (WTVs); tumor subcompartments of necrosis, gadolinium enhancement, and nonenhancing T2 (NCE T2) hyperintensity; and normal-appearing white matter (NAWM), and pH_i values were compared. Spearman correlation was used to assess association between pH_i and the proliferation index Ki-67. For all study participants, mean pH_i values were higher in the WTV (7.057 ± 0.024) compared with NAWM (7.006 ± 0.012 ; $P < .001$). In eight participants with high-grade gliomas, pH_i was increased in all tumor subcompartments (necrosis, 7.075 ± 0.033 ; gadolinium enhancement, 7.075 ± 0.024 ; NCE T2 hyperintensity, 7.043 ± 0.015) compared with NAWM (7.004 ± 0.014 ; all $P < .01$). The pH_i values of WTV positively correlated with Ki-67 ($R^2 = 0.74$, $r = 0.78$, $P = .001$). In conclusion, ^{31}P spectroscopic MRI at 7.0 T enabled high-resolution quantification of pH_i in gliomas, with pH_i alteration associated with the Ki-67 proliferation index, and may aid in diagnosis and treatment monitoring.

Supplemental material is available for this article.

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Neoplastic cells exhibit major metabolic and microenvironmental transformations with distinct changes in pH (1). Whereas the extracellular pH becomes more acidic, the intracellular pH (pH_i) shifts to the alkaline side, demonstrating a reversed pH gradient compared with physiologic conditions (2). The reversed pH gradient has been reported to stimulate tumor invasiveness, as the acidic extracellular pH enhances tumor neoangiogenesis (3,4) and the alkaline pH_i reduces apoptosis and enhances proliferation (5,6).

To map pH changes, various MRI techniques have evolved. Such techniques include chemical exchange saturation transfer (7–9), which relies on the assessment of endogenous metabolites with limited specificity (10), and phosphorous ^{31}P spectroscopic MRI, which can characterize cellular bioenergetics with high specificity but limited spatial resolution (11,12). Since the chemical shifts of ^{31}P metabolites are strongly dependent on pH, pH_i values can be quantified from the chemical shift difference between intracellular inorganic phosphate (P_i) and phosphocreatine resonance, respectively. Recent investigations at ultrahigh magnetic fields (≥ 7.0 T) enabled high-resolution three-dimensional (3D) pH mapping (effective voxel volume, 5.7 mL) and demonstrated the existence of a compartmentalized P_i resonance

in the human brain, displayed as a downfield peak in reference to the well-known intracellular P_i resonance in the ^{31}P spectrum (13). The identification and subsequent extraction of this second P_i pool has demonstrated advantages in more reliable and robust determination of pH_i values (14).

The purpose of this study was to investigate pH_i alterations in a study sample of participants with newly diagnosed glioma using a high-resolution ^{31}P spectroscopic MRI approach at 7.0 T. We hypothesized that 3D ^{31}P spectroscopic MRI allows for the quantification of increased pH_i in gliomas and assessment of pH_i heterogeneity in brain tumors of different histologic and genetic subtypes. Compared with previously published ^{31}P spectroscopic MRI studies (15,16), this study aims to assess pH_i differences for distinct tumor subvolumes with improved specificity by using comparably smaller voxel volumes and removing confounding contributions to P_i .

Materials and Methods

Study Participants and Design

This prospective study was conducted between September 2018 and November 2019 and included 13 partici-

Abbreviations

IDH = isocitrate dehydrogenase, MGMT = O⁶-methylguanine-DNA-methyltransferase, NAWM = normal-appearing white matter, NCE T2 = nonenhancing T2, pH_i = intracellular pH, P_i = (intracellular) inorganic phosphate, 3D = three-dimensional, WHO = World Health Organization, WTV = whole-tumor volume

Summary

Phosphorous 31 spectroscopic MRI at 7.0 T enabled high-resolution quantification of increased intracellular pH in the tumor volume, which was associated with histologic features, in study participants with glioma.

Key Points

- Mean intracellular pH (pH_i), quantified using phosphorous 31 7.0-T spectroscopic MRI, was higher in the whole-tumor volume compared with normal-appearing white matter in all study participants with newly diagnosed glioma (7.057 ± 0.024 [SD] vs 7.006 ± 0.012 , $P < .001$).
- The pH_i varied between investigated tissue subvolumes ($P < .001$): pH_i was increased in all tumor subcompartments of high-grade gliomas versus normal-appearing white matter (necrosis, 7.075 ± 0.033 ; gadolinium enhancement, 7.075 ± 0.024 ; nonenhancing T2 hyperintensity, 7.043 ± 0.015 vs 7.004 ± 0.014 ; all $P < .01$).
- The pH_i in the whole-tumor volume correlated positively with the proliferation index Ki-67 ($R^2 = 0.74$, $r = 0.78$, $P = .001$).

Keywords

³¹P MRSI, pH, Glioma, Glioblastoma, Ultra-High-Field MRI, Imaging Biomarker, 7 Tesla

patients with MRI-observable brain lesions suspicious for glioma. Study approval was given by the local institutional review board. Written informed consent was obtained from all study participants prior to study inclusion. Criteria for eligibility were age of at least 18 years, newly diagnosed lesion suspicious for glioma, and 7.0-T MRI eligibility. Exclusion criteria were age less than 18 years and ineligibility for 7.0-T MRI. All participants underwent clinical routine MRI scans at 3.0 T (Department of Neuroradiology, Heidelberg University Hospital, Heidelberg, Germany), with an average of 5.67 days (range, 0–17 days) elapsing between the clinical routine and 7.0-T research scans. Because of inadequate data quality, one study participant was excluded from further analyses, resulting in a total of 12 study participants (Fig 1). In the excluded participant, B₀ field inhomogeneities at the frontobasis (in direct proximity to the paranasal sinuses) caused severe line broadening in the spectra, hindering a reliable separation of the extracellular P_i/P_i resonance with ³¹P spectroscopic MRI. Findings in seven study participants have previously been reported in technical development publications (14,17). These prior articles were focused on new technical developments, whereas in this article, we report on clinical aspects (ie, pH_i alterations in tumors related to different histologic analyses and genetic subtypes).

7.0-T MRI Protocol and Data Analysis

The ³¹P spectroscopic MRI methodical approach was implemented with a 7.0-T whole-body scanner (Magnetom 7T; Siemens Healthineers), using a double-resonant ³¹P/¹H phased-array head coil with 32 ³¹P receiver elements (Rapid Biomedical)

for improved signal-to-noise ratio performance. The MRI protocol and data analysis are available in Appendix S1.

Volumetric segmentations of the whole-tumor volume, central necrosis, nonenhancing T2 hyperintensity, gadolinium enhancement, and normal-appearing white matter (NAWM) were performed by two radiologists in consensus (D.P., K.D.H.; 10 and 7 years of experience in neuroimaging, respectively; Fig S1). The subvolume of NAWM was delineated in the contralateral hemisphere. NAWM delineation was performed by the same two readers to test for interreader variability using Bland-Altman analysis. For study participant 12 with a bifrontal tumor location, the NAWM volume was selected in the contralateral parietal lobe.

The tumor size was assessed by means of the largest lesion diameter of gadolinium enhancement on T1-weighted magnetization-prepared rapid-acquisition gradient-echo images for all participants with high-grade gliomas and on T2-weighted fluid-attenuated inversion recovery images for nonenhancing low-grade gliomas.

Statistical Analysis

The pH_i values were compared between the whole-tumor volume and NAWM over all study participants using Wilcoxon signed rank tests for paired samples. The Wilcoxon rank sum test was used to test for differences in pH_i between high- and low-grade gliomas, isocitrate dehydrogenase (IDH)-wildtype and IDH-mutant gliomas, and between O⁶ methylguanine-DNA-methyltransferase (MGMT)-methylated and MGMT-unmethylated glioma subgroups. The Spearman rank order correlation test (represented by the Spearman rank order coefficient [r_s]) and the determination coefficient (R^2) were used to assess the correlation between pH_i and the proliferation index Ki-67, participant age, and tumor size. In participants with high-grade gliomas, pH_i differences in tumor subvolumes were compared by using analysis of variance after normality testing passed, followed by post hoc pairwise comparisons (Holm-Sidak method). The significance level was set to $P < .05$. For the Wilcoxon rank sum tests for World Health Organization (WHO) grade, IDH status, and MGMT status, significance level was set to $P < .05/3 = .017$ (Bonferroni correction). All statistical analyses were performed with SigmaPlot (version 14.0; Systat Software).

Results

Participant Characteristics

Twelve participants (mean age, 58 years $\pm$ 18 [SD]; seven male, five female) with newly diagnosed previously untreated glioma were analyzed. Characteristics of all study participants and histopathologic results are shown in Table 1.

Intracellular pH Analysis in Glioma

The pH_i was increased in whole-tumor volume (7.057 ± 0.024) compared with that in NAWM (7.006 ± 0.012) in all study participants with glioma ($P < .001$) (Fig 2), which is demonstrated in corresponding maps of two participants

with glioblastoma (WHO grade IV; Fig 3A, 3B) and one participant with low-grade glioma (WHO grade II, astrocytoma; Fig 3C). Table 2 lists the complete results for the entire study cohort. Regarding intrareader dependencies, no evidence of differences was found between the readers for NAWM segmentation (Table S1), as further demonstrated by Bland-Altman analysis (Fig S2).

A positive correlation was found between the proliferation index Ki-67 and pH_i in the whole-tumor volume ($R^2 = 0.74$, r_s

$= 0.78$, $P = .001$), while no correlation was observed between Ki-67 and pH_i in NAWM ($R^2 = 0.01$, $r_s = 0.17$, $P = .59$) (Fig 4A). Age and pH_i showed positive correlation in the whole-tumor volume ($R^2 = 0.32$, $r_s = 0.60$, $P = .04$). No correlation was observed between age and pH_i in NAWM ($R^2 = 0.08$, $r_s = -0.01$, $P = .97$) (Fig 4B) or between maximum tumor diameter and pH_i (Fig 4C). No evidence of a difference in pH_i was found between high-grade tumors (7.065 ± 0.025) and low-grade tumors (7.041 ± 0.013 , $P = .15$) (Fig S3A). Similarly, regarding *IDH1* mutation and *MGMT* methylation, there was no evidence of a difference in pH_i between the groups (*IDH*-wildtype = 7.066 ± 0.024 vs *IDH*-mutant = 7.038 ± 0.009 , $P = .048$, Fig S3B; *MGMT* methylated = 7.064 ± 0.027 vs *MGMT* unmethylated = 7.054 ± 0.021 , $P = .792$, Fig S3C).

Intracellular pH Analysis in Tumor Subvolumes of Interest in High-Grade Glioma

In participants with high-grade gliomas ($n = 8$), pH_i varied between the investigated tissue subvolumes ($P < .001$). The pH_i values of all subcompartments were increased (necrosis, 7.075 ± 0.033 ; gadolinium enhancement, 7.075 ± 0.024 ; nonenhancing T2 hyperintensity, 7.043 ± 0.015) compared with NAWM (7.004 ± 0.014) (all $P < .01$) (Fig 2). Increased pH_i values were observed between necrosis and gadolinium enhancement compared with nonenhancing T2 hyperintensity ($P = .02$ and $P = .02$). No evidence of differences was found between the other tumor subcompartments.

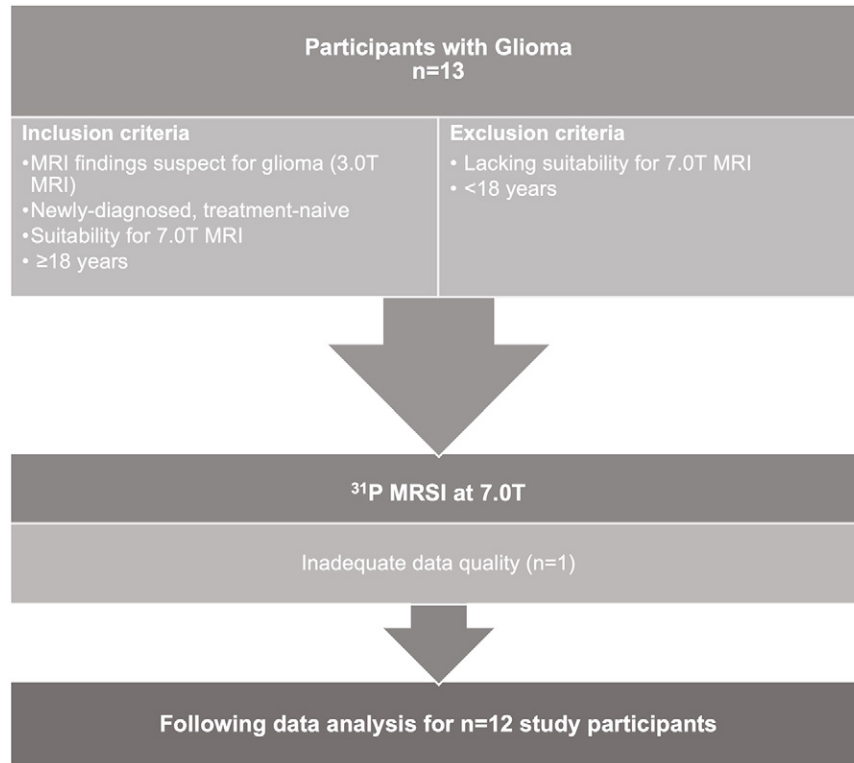


Figure 1: Flowchart of study inclusion and exclusion criteria. MRSI = MR spectroscopic imaging, ^{31}P = phosphorous 31.

Table 1: Characteristics of Study Participants

Participant No.	Age (y)	Sex	WHO Grade	Histopathology	<i>IDH</i> Status	Tumor Location	Largest Diameter (mm)
1	71	Male	IV	Glioblastoma	WT	Right temporal/parietal	29.46
2	57	Male	IV	Glioblastoma	WT	Left temporal/occipital	39.55
3	68	Male	IV	Glioblastoma	WT	Right temporal/parietal	64.85
4	78	Female	IV	Glioblastoma	WT	Left temporal	47.99
5	36	Female	III	Astrocytoma	MUT	Right frontal	46.31
6	67	Female	IV	Glioblastoma	WT	Right temporal	45.32
7	80	Female	IV	Glioblastoma	WT	Left temporal	40.72
8	75	Male	IV	Glioblastoma	WT	Left occipital	61.05
9	57	Female	II	Astrocytoma	MUT	Left frontal	75.01
10	34	Male	II	Astrocytoma	MUT	Left temporal	47.09
11	23	Male	II	Astrocytoma	MUT	Right frontal	48.91
12	50	Male	II	Pleomorphic glioma	WT	Bifrontal	87.17

Note.—All study participants had histopathologically proven glioma. MUT = mutated, WHO = World Health Organization, WT = wildtype.

Figure 2: Region-specific assessment of intracellular pH (pH_i) alterations over all study participants with glioma ($n = 12$). Box plots of mean pH_i values in normal-appearing white matter (NAWM), whole-tumor volume (WTV), and subvolumes of interest (necrosis, gadolinium enhancement [GDCE], nonenhancing T2 hyperintensity [NCE T2]) in all participants with high-grade glioma (HGG) (World Health Organization grade III–IV, $n = 8$). The pH_i of the WTV was increased compared with that of NAWM in all participants ($P < .001$). Analysis of tumor subregions yielded increased mean pH_i values for subvolumes of interest compared with NAWM ($P < .01$). The pH_i in necrosis and GDCE was increased compared with NCE T2 (necrosis, $P = .02$; GDCE, $P = .02$), with no evidence of differences between the other tumor subcompartments. Mean pH_i values for all study participants and subvolumes are shown (gray circles). The red line in the middle of each box is the sample median. The bottom and top of each box are the 25th and 75th percentiles of the sample, respectively. The distance between the bottom and top of each box is the IQR. The whiskers of the box plots extend from the ends of the box to the smallest and largest data values.

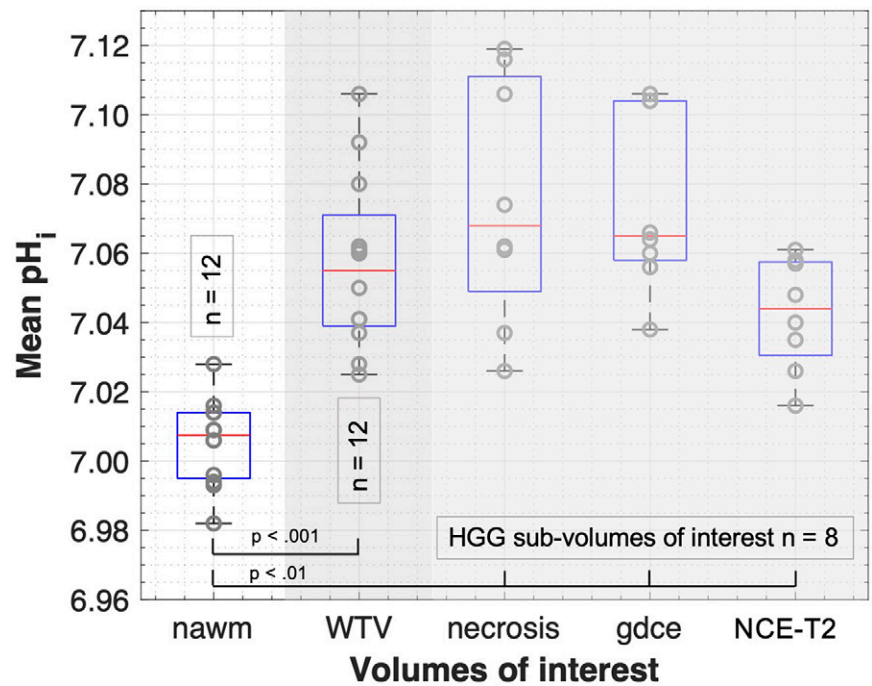


Table 2: Summary of pH_i Values of All Study Participants with Glioma

Glioma Type, Participant No., and Measurement	Tumor Subvolumes of Interest				
	NAWM	WTV	Necrosis	GDCE	NCE T2
HGG ($n = 8$)					
1	7.009 $\pm$ 0.004	7.060 $\pm$ 0.008	7.061 $\pm$ 0.007	7.060 $\pm$ 0.008	7.061 $\pm$ 0.009
2	6.982 $\pm$ 0.003	7.028 $\pm$ 0.035	7.037 $\pm$ 0.034	7.038 $\pm$ 0.040	7.016 $\pm$ 0.025
3	6.993 $\pm$ 0.004	7.080 $\pm$ 0.059	7.106 $\pm$ 0.049	7.104 $\pm$ 0.054	7.057 $\pm$ 0.056
4	7.014 $\pm$ 0.001	7.092 $\pm$ 0.050	7.119 $\pm$ 0.040	7.104 $\pm$ 0.047	7.040 $\pm$ 0.013
5	7.028 $\pm$ 0.002	7.050 $\pm$ 0.024	7.026 $\pm$ 0.020	7.056 $\pm$ 0.025	7.048 $\pm$ 0.023
6	6.994 $\pm$ 0.005	7.041 $\pm$ 0.033	7.062 $\pm$ 0.031	7.066 $\pm$ 0.042	7.026 $\pm$ 0.017
7	7.016 $\pm$ 0.004	7.061 $\pm$ 0.026	7.074 $\pm$ 0.012	7.064 $\pm$ 0.024	7.058 $\pm$ 0.028
8	6.996 $\pm$ 0.002	7.106 $\pm$ 0.045	7.116 $\pm$ 0.046	7.106 $\pm$ 0.042	7.035 $\pm$ 0.012
Mean	7.004 $\pm$ 0.014	7.065 $\pm$ 0.025	7.075 $\pm$ 0.033	7.075 $\pm$ 0.024	7.043 $\pm$ 0.015
Median	7.003 (0.022)	7.061 (0.046)	7.068 (0.071)	7.065 (0.047)	7.044 (0.030)
LGG ($n = 4$)					
9	7.009 $\pm$ 0.001	7.025 $\pm$ 0.012			
10	7.006 $\pm$ 0.002	7.037 $\pm$ 0.049			
11	7.014 $\pm$ 0.003	7.041 $\pm$ 0.028			
12	7.006 $\pm$ 0.004	7.062 $\pm$ 0.051		7.100 $\pm$ 0.052	7.052 $\pm$ 0.045
Mean	7.009 $\pm$ 0.003	7.041 $\pm$ 0.013			
Median	7.008 (0.007)	7.039 (0.029)			
All ($n = 12$)					
Mean	7.006 $\pm$ 0.012	7.057 $\pm$ 0.024			
Median	7.008 (0.019)	7.055 (0.037)			

Note.—Data are intracellular pH (pH_i) values and are presented as either means $\pm$ SDs or medians and IQRs (in parentheses) in the whole-tumor volume (WTV) and normal-appearing white matter (NAWM) in all study participants with glioma. In the subcohort analysis of participants with high-grade glioma (HGG) ($n = 8$), the whole-tumor volume was further segmented into three subvolumes of interest: gadolinium enhancement (GDCE), necrosis, and nonenhancing T2 hyperintensity (NCE T2). LGG = low-grade glioma.

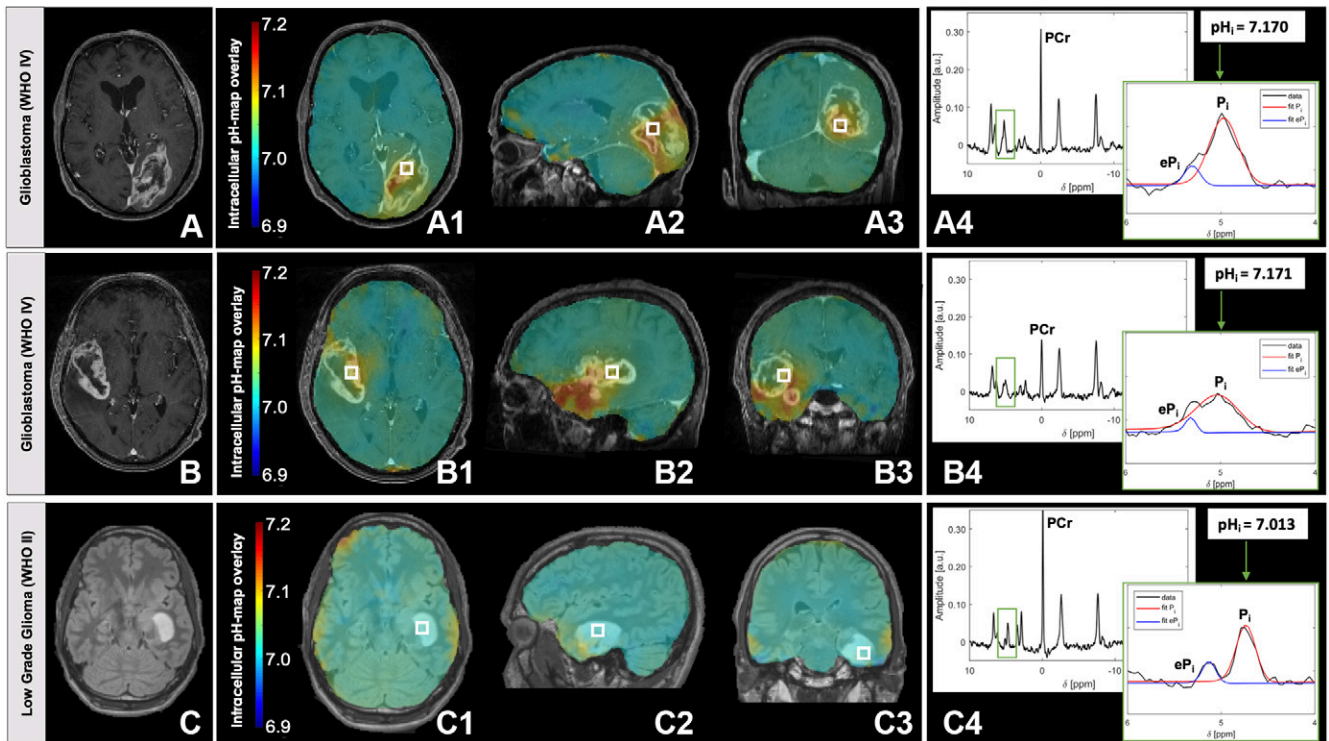


Figure 3: Whole-brain intracellular pH (pH_i) images in participants with high- and low-grade glioma obtained using phosphorous ^{31}P 7.0-T spectroscopic MRI. **(A, B)** Clinical (3.0 T) contrast-enhanced T1-weighted magnetization-prepared rapid-acquisition gradient-echo images with overlays of the volumetric three-dimensional (3D) color-coded pH_i map in all three planes in a 75-year-old man (**A**, study participant 8 in Table 1) and a 68-year-old man (**B**, study participant 3 in Table 1), both with histopathologically proven glioblastoma (World Health Organization [WHO] grade IV). Axial (**A1, B1**), sagittal (**A2, B2**), and coronal (**A3, B3**) planes show an alkaline shift of pH_i in the whole-tumor volume (**A**, $pH_i = 7.106 \pm 0.045$; **B**, $pH_i = 7.080 \pm 0.059$) compared with normal-appearing white matter (**A**, $pH_i = 6.996 \pm 0.002$; **B**, $pH_i = 6.993 \pm 0.004$). **(C)** Clinical (3.0 T) fluid-attenuated inversion recovery images with overlays of the volumetric 3D color-coded pH_i maps in a 34-year-old man (study participant 10 in Table 1) with histopathologically proven astrocytoma (WHO grade II) in axial (**C1**), sagittal (**C2**), and coronal (**C3**) planes. No pronounced pH_i alterations were observed in the tumor region (**C**). Highly resolved ^{31}P spectra with corresponding pH_i values for the indicated voxels (white box) are shown for all participants (**A4, B4, C4**) with respect to both inorganic phosphate peaks (P_i , eP_i) for pH_i quantifications. eP_i = extracellular inorganic phosphate, P_i = intracellular inorganic phosphate, PCr = phosphocreatine.

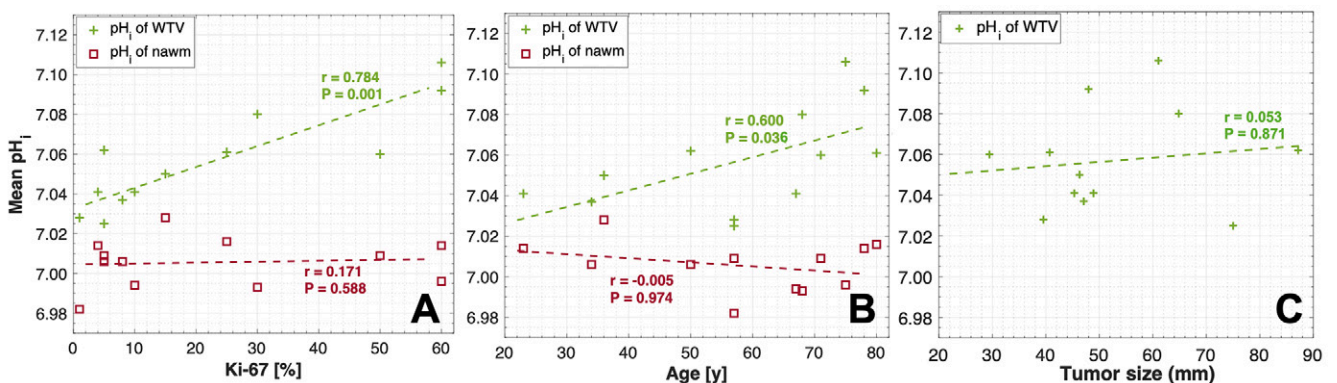


Figure 4: Spearman correlation of mean intracellular pH (pH_i) in the whole-tumor volume (WTV) and normal-appearing white matter (NAWM) with respect to **(A)** proliferation index Ki-67, **(B)** age, and **(C)** tumor size for all study participants with glioma ($n = 12$). **(A)** Spearman correlation analysis of mean pH_i in NAWM and WTV with proliferation index Ki-67 showed no correlation for pH_i in NAWM ($R^2 = 0.01$, $r_s = 0.17$, $P = .59$) and a positive correlation for the pH_i within the WTV ($R^2 = 0.74$, $r_s = 0.78$, $P = .001$). **(B)** Spearman correlation analysis of mean pH_i in NAWM and WTV with age yielded no correlation for NAWM ($R^2 = 0.08$, $r_s = -0.01$, $P = .97$) and a positive correlation for WTV ($R^2 = 0.32$, $r_s = 0.60$, $P = .04$). **(C)** Spearman correlation analysis of mean pH_i of the WTV with tumor size showed no correlation ($R^2 = 0.004$, $r_s = 0.05$, $P = .87$).

Discussion

This work shows that noninvasive high-resolution mapping of pH_i is feasible using ^{31}P spectroscopic MRI at 7.0 T, as demonstrated in a study cohort of participants with previously untreated glioma. The relatively small voxel size of 5.7 mL en-

abled high-resolution 3D pH_i mapping, showing an association of pH_i with the Ki-67 proliferation index in participants with glioma.

The pH_i was increased in the whole-tumor volume compared with NAWM in all study participants with newly diagnosed

glioma. Mean pH_i values for NAWM obtained in this study are consistent with those in previous ^{31}P spectroscopic MRI studies performed at 1.5 T and 3.0 T with comparably lower resolution (18,19). Schüre et al investigated tumor pH_i in patients with newly diagnosed glioblastoma (18), and Maintz et al reported similar pH_i values obtained at 1.5-T MRI in patients with low- or high-grade gliomas (19). The difference in voxel size, and thus the resolution, of ^{31}P spectroscopic MRI achieved at 7.0 T is more than one order of magnitude compared with the 1.5- and 3.0-T MRI approaches with voxel sizes ranging from 56 to 129 mL (19), yielding reduced partial volume effects in our study. To our knowledge, this is the first study to evaluate pH_i values in specific 3D-segmented tumor subvolumes. This was feasible due to the comparatively high resolution of the used approach and the large tumors of the participants enrolled in this study.

The pH_i in the whole-tumor volume positively correlated with the proliferation index Ki-67. This association may support use of pH_i as an imaging biomarker for prognostication, as Ki-67 has been demonstrated as a prognostic factor in patients with glioma (20). Previous in vivo studies investigated pH_i mapping with ^{31}P spectroscopic MRI as an imaging biomarker for tumor progression and for the evaluation of therapy response in patients with recurrent glioblastoma (16). Since tumor relapse is associated with altered metabolic activity in cancer tissue, detectable changes in pH_i might aid in the differentiation of tumor progression from radiation-related effects (ie, radionecrosis). ^{31}P spectroscopic MRI could also aid therapy response monitoring for other therapeutic strategies, such as treatment with temozolomide (21) or ion transporter inhibitors (22).

Our study demonstrated no evidence of a difference in pH_i between high- and low-grade gliomas, while Wenger et al (23) observed increased pH_i in high-grade tumors (WHO III–IV) compared with low-grade gliomas. We also found no evidence of a difference in pH_i with respect to *IDH1* status. Although there was a pronounced variation in pH_i between both groups, it lacked statistical significance. Schüre et al (18) found increased pH_i values for *IDH*-wildtype tumors; in a larger study sample, this finding might also be reproducible with the approach used in this work. Other techniques that have been reported as potential MRI biomarkers to assess *IDH* mutation include the detection of 2-hydroxyglutarate via ^1H MR spectroscopy (24), chemical exchange saturation transfer imaging (25,26), diffusion- and perfusion-weighted MRI approaches (27), as well as radiomics and deep learning-based approaches using conventional MRI data (28,29). However, these approaches still require independent external validation studies to enable broader clinical use.

Forester et al (31) reported reduced pH_i with increasing age in healthy individuals. In this study, pH_i alteration in the whole-tumor volume and age showed a strong positive association.

Regarding pH_i variations in the healthy brain, it should be noted that a certain pH variability, depending on age and condition, is physiologic. These pH_i heterogeneities are also observed between white and gray matter tissues (14,30). Note that since the spatial resolution is still limited with this imaging technique, partial volume effects (ie, tissue mixing) further contribute to the observed variations. Furthermore, technical inaccuracies,

primarily due to varying signal-to-noise ratio in different voxels, can lead to imprecisions of the pH quantification results.

This study had some limitations. First, the study had a relatively small sample size ($n = 12$). Differences in pH_i , especially with respect to WHO tumor grade and *IDH1* and *MGMT* status, must be interpreted with care since the investigated subcohorts have even smaller sample sizes. Respectively, factors such as age and other demographics could have impacted our findings, highlighting the need to investigate larger study cohorts in the future. Second, regarding the analysis of tumor subvolumes in high-grade glioma, partial volume effects might diminish the actual pH_i difference of the defined volumes, as the 5.7-mL voxels are still comparatively large. Finally, a total measurement time as long as 65 minutes for the entire scan protocol used in this study is not feasible in clinical routine. However, simulations showed that the scan time for ^{31}P spectroscopic MRI can be reduced to 20 minutes via low-rank denoising (14). To further improve clinical applicability, it would be desirable to perform only one clinically adapted 7.0-T scan, including an exchange of head coils, instead of the two-scanner approach performed in this study. Ultimately, the availability of human 7.0-T MRI scanners is still limited, and additional equipment is required for ^{31}P spectroscopic MRI, including dedicated double-resonant ($^{31}\text{P}/^1\text{H}$) coils. This work has been performed as a single-center study, using the same 7.0-T MRI scanner.

In conclusion, this work showed that high-resolution pH_i mapping and robust pH_i quantification via whole human brain ^{31}P spectroscopic MRI is feasible, as demonstrated in a cohort of participants newly diagnosed with glioma. ^{31}P spectroscopic MRI provides insights into metabolic profiles and local pH_i heterogeneity in glioma tumors that could aid initial diagnosis and treatment monitoring. Further optimization, especially acceleration of scan time, needs to be realized to translate this method into clinical use and patient benefit.

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